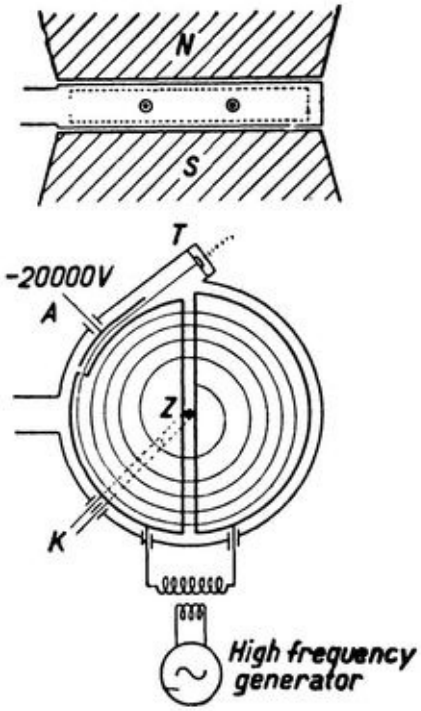


Figure 37.—Cyclotron.

The adjustment of such an apparatus calls for a great deal of technical skill. Moreover, the cyclotron is a machine of such dimensions as are seldom encountered in any other appliance used in physical research. Let us illustrate this by mentioning a few figures. The pole pieces of a cyclotron in use in the United States for some time are 95 cm. in diameter. The magnet of this cyclotron produces a magnetic field of 14,000 oersteds, and 60 tons of iron and 10 tons of copper went into its construction. The production of the magnetic field of 14,000 oersteds requires an input of 30 kilowatts. If this cyclotron is used to accelerate deuterons, these will emerge from it with an energy of 9 Mev.—in other words, as if they had passed through a potential drop of 9,000,000 volts. A current of only 0·1 milliampere flowing through this drop of potential would represent a power of nearly 1 kilowatt (900 watts to be precise). As each particle carries an elementary quantum of electricity, $1{\cdot}6 \times 10^{-19}$ coulombs, it is easy to compute that this current is the equivalent of roughly 6×10^{14} particles per second.

Figure 38 shows an exterior view of such a cyclotron. The windings of the magnet are visible; between the pole pieces of the magnet are the dees, in which the particles begin their acceleration. The beam emerging from the cyclotron is also visible in the photograph.

Numerous cyclotrons are already in use in the United States. Several have been built in Europe, too. Germany has had one since 1944, in the *Kaiser Wilhelm Institut* in Heidelberg, destined primarily for medical use. The universal recognition of the importance of the cyclotron is demonstrated by